

Blood Vessel Enhancement and Segmentation Using Wavelet Transform

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Abstract—Retinal vessel segmentation is an essential step for the diagnoses of various eye diseases. An automated tool for blood vessel segmentation is useful to eye specialists for purpose of patient screening and clinical study. Vascular pattern is normally not visible in retinal images. In this paper, we present a method for enhancing, locating and segmenting blood vessels in images of retina. We present a method that uses 2-D Gabor wavelet and sharpening filter to enhance and sharpen the vascular pattern respectively. This technique locates and segments the blood vessels using edge detection algorithm and morphological operations. This technique is tested on publicly available STARE database of manually labeled images which has been established to facilitate comparative studies on segmentation of blood vessels in retinal images. The validation of our retinal image vessel segmentation technique is supported by experimental results.

Keywords—Vascular pattern, retinal images, vessel segmentation

I. INTRODUCTION

Retinal image processing is greatly required in diagnosing and treatment of many diseases affecting the retina and the choroid behind it [1]. Diabetic retinopathy is one of the complications of diabetes mellitus affecting the retina and the choroid. In this condition, a network of small blood vessels, called choroidal neovascularization (CNV), arises in the choroid and taking a portion of the blood supplying the retina. As the amount of blood supplying the retina is decreased, the sight may be degraded and in the severe cases, blindness may occur. The eye is a window to the retinal vascular system which is uniquely accessible for the study of a continuous vascular bed in humans [1].

The detection and measurement of blood vessels can be used to quantify the severity of disease, as part of the process of automated diagnosis of disease or in the assessment of the progression of therapy [2]. Retinal blood vessels have been shown to have measurable changes in diameter, branching angles, length, as a result of a disease.

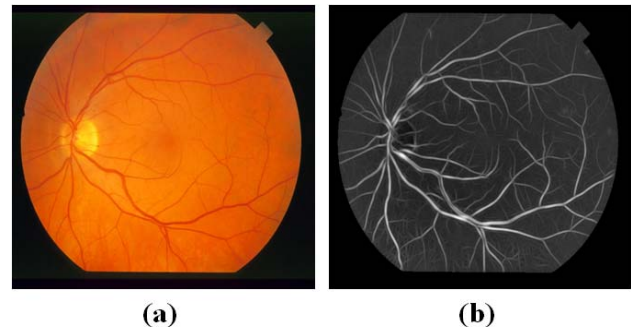


Figure 1. (a) Original Retinal Image (b) Enhanced Vascular Pattern

Thus a reliable method of vessel segmentation would be valuable for the early detection and characterization of changes due to such diseases [2]. Retinal vascular pattern facilitates the physicians for the purposes of diagnosing eye diseases, patient screening, and clinical study [3]. Inspection of blood vessels provides the information regarding pathological changes caused by ocular diseases including diabetes, hypertension, stroke and arteriosclerosis [4]. The hand mapping of retinal vasculature is a time consuming process that entails training and skill. Automated segmentation provides consistency and reduces the time required by a physician or a skilled technician for manual labeling [5].

Retinal vessel segmentation may be used for automatic generation of retinal maps for the treatment of age-related macular degeneration [6], extraction of characteristic points of the retinal vasculature for temporal or multimodal image registration [7], retinal image mosaic synthesis, identification of the optic disc position [8], and localization of the fovea [9]. The challenges faced in automated vessel detection include wide range of vessel widths, low contrast with respect with background and appearance of variety of structures in the image including the optic disc, the retinal boundary and other pathologies [10]. Fig. 1 shows the

retinal image taken from STARE database and the enhanced vascular pattern [11].

Different approaches for automated vessel segmentation have been proposed. Methods based on vessel tracking to obtain the vasculature structure, along with vessel diameters and branching points have been proposed by [12]-[17]. Tracking consists of following vessel center lines guided by local information. In [23], ridge detection was used to form line elements and partition the image into patches belonging to each line element. Pixel features were then generated based on this representation. Many features were presented and a feature selection scheme is used to select those which provide the best class separability. Papers [18]-[21] used deformable models for vessels segmentation. Chuadhuri et al. [22] proposed a technique using matched filters to emphasize blood vessels. An improved region based threshold probing of the matched filter response technique was used by Hoover et al. [24].

In this paper, we present the colored retinal image vessel segmentation technique that enhances and sharpens the vascular pattern using 2-D Gabor wavelet and sharpening filters. Our technique creates a binary mask for vessel segmentation applying edge detection algorithm on sharpened retinal image and a fine segmentation mask is obtained by applying morphological dilation operation.

The paper is organized in four sections. In Section II, a schematic overview of our implementation methodology is illustrated. Section II also presents the step by step techniques required for colored retinal image vessels segmentation. Experimental results of the tests on the images of the STARE database and their analysis are given in Section III followed by conclusion in Section IV.

II. BLOOD VESSEL ENHANCEMENT AND SEGMENTATION

A schematic overview of proposed blood vessel enhancement and segmentation method is illustrated here. The monochromatic RGB retinal image is taken as an input and 2-D Gabor wavelet is used to enhance the vascular pattern especially the thin and less visible vessels are enhanced using Gabor wavelet [25]. Before extracting vessels from enhanced retinal image, the blood vessels are sharpened using sharpening filter [26].

Vessels segmentation binary mask is created by detecting vessels edges from sharpened image. The blood vessels are marked by the masking procedure which assign one to all those pixels which belong to blood vessels and zero to non vessels pixels. Final refined vessel segmentation mask is created by applying morphological dilation operator [26].

Fig. 2 shows the complete flow diagram of proposed blood vessel enhancement and segmentation technique.

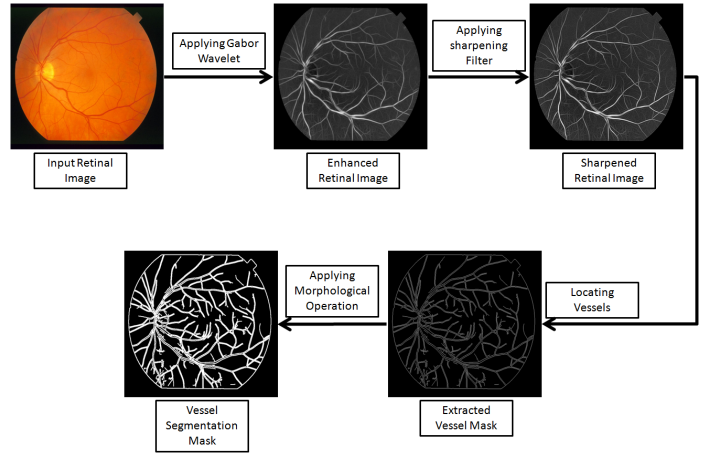


Figure 2. Complete flow diagram for vessel enhancement and segmentation

A. Vessel Enhancement

We have used 2-D Gabor wavelet to enhance the vascular pattern and thin vessels [25]. 2-D Gabor wavelet is used due to its directional selectiveness capability of detecting oriented features and fine tuning to specific frequencies [25], [27].

- The continuous wavelet transform $T_\psi(\mathbf{b}, \theta, a)$ is defined in terms of the scalar product of f with the transformed wavelet $\Psi_{\mathbf{b}, \theta, a}$ using (1) [28]

$$T_\psi(\mathbf{b}, \theta, a) = C_\psi^{-1/2} \langle \psi_{\mathbf{b}, \theta, a} | f \rangle$$

$$= C_\psi^{-1/2} a^{-1} \int \psi^*(a^{-1} r_{-\theta}(\mathbf{x} - \mathbf{b})) f(\mathbf{x}) d^2 \mathbf{x} \quad (1)$$

where $f \in L^2$ is an image represented as a square integrable i.e., finite energy) function defined over $\mathbb{R}^2$ and $\Psi \in L^2$ be the analyzing wavelet. C_ψ , Ψ , $\mathbf{b}$, θ , and a denote the normalizing constant, analyzing wavelet, the displacement vector, the rotation angle, and the dilation parameter respectively.

- It is easy to implemented wavelet transform using the fast Fourier transform algorithm. Fourier wavelet transform is defined using (2) [27].

$$T_\psi(\mathbf{b}, \theta, a) = C_\psi^{-1/2} a \int \exp(j\mathbf{k}\mathbf{b}) \hat{\psi}^*(a\mathbf{r}_{-\theta}\mathbf{k}) \hat{f}(\mathbf{k}) d^2 \mathbf{k} \quad (2)$$

where $j = \sqrt{-1}$, and the hat ($\hat{\psi}^*$ and $\hat{f}$) denotes a Fourier transform.

- The 2-D Gabor wavelet is defined as [28]

$$\psi_G(\mathbf{x}) = \exp(j\mathbf{k}_0\mathbf{x}) \exp(-\frac{1}{2}|\mathbf{A}\mathbf{x}|^2) \quad (3)$$

where $\mathbf{k}_0$ is a vector that defines the frequency of the complex exponential and $\mathbf{A} = \text{diag}[\epsilon^{-1/2}, 1]$, $\epsilon \geq 1$ is a 2×2 diagonal matrix that defines the elongation of filter in any desired direction.

- For each pixel position and considered scale value,

$$M_\psi(\mathbf{b}, a) = \max_\theta |T_\psi(\mathbf{b}, \theta, a)| \quad (4)$$

the Gabor wavelet transform $M_\psi(\mathbf{b}, a)$ is computed using (4) [27], for θ spanning from 0° up to 170° at steps of 10° and the maximum is taken.

Fig. 3 shows the enhanced vascular pattern after applying Gabor wavelet. Fig. 3(b) shows the enhanced vascular pattern using Hoover et al. method [24] while fig. 3(c) shows the vascular pattern enhanced using wavelet method and there is a clear improvement in enhancement using wavelet method.

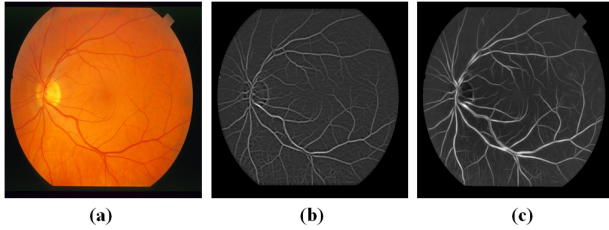


Figure 3. (a) Original Retinal Image; (b) Enhanced Vascular Pattern using Hoover et. al method[24]; (c) Enhanced Vascular Pattern using Gabor Wavelet

B. Sharpening Filter

We used unsharp masking [26] on enhanced vessel retinal image to sharpen the vascular pattern. The application of Gabor wavelet on colored retinal image enhances the vascular pattern but the resulting image is a little blurred so we have used unsharp filter to sharpen the vascular edges. This helps in reliable extraction of vessels from the colored retinal image.

Fig. 4(a) shows enhanced retinal image and fig. 4(b) shows sharpened retinal image. It is clearly visible that the vessels are much more prominent than they were in the original image.

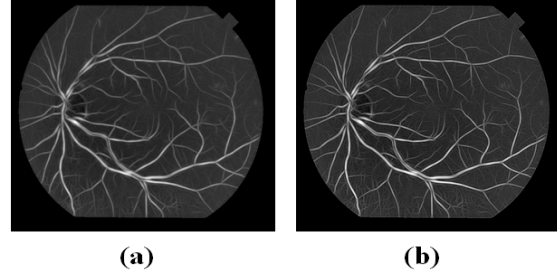


Figure 4. (a) Enhanced Vascular Pattern; (b) Sharpened Vascular Pattern

C. Vessel Segmentation Mask

Vessels segmentation mask is created by extracting vessels boundaries using edge detection algorithm and then by applying morphological dilation operator [26]. In this paper, we have used canny edge detector [29]. Canny operator simultaneously optimizes three criteria: detection criterion, localization criterion, and elimination of multiple responses and these rules also become a standard to evaluate edge detection algorithm performance [29].

The vessels extracted using edge detection algorithm are refined using standard morphological dilation operator [26]. Dilation will fill the gaps between vessel boundaries detected by edge detection algorithm.

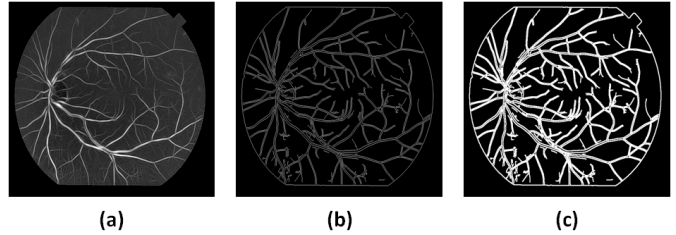


Figure 4. (a) Sharpened Retinal Image; (b) Extracted Vessels; (c) Morphological Refined Vessels

Fig. 5(b) shows the hollow vessels extracted using edge detection algorithm and fig. 5(c) shows the result of applying morphological dilation on 5(b).

III. EXPERIMENTAL RESULTS

The tests of proposed technique are performed with respect to the vessel segmentation accuracy and their standard deviation using publicly available STARE database [11]. The STARE database consists of 20 RGB color images of the retina. The images are of size 605×700 pixels, 24 bits per pixel (standard RGB). There are two hand-labeling available for the 20 images of made by two different human observers.

The manually segmented images by 1st human observer are used as ground truth and the segmentations of set B are

tested against set A, serving as a human observer reference for performance comparison truth. The segmentations of set B are tested against those of A, serving as a human observer reference for performance comparison [24]. We compared the accuracy of proposed technique with the accuracies of the methods of Hoover et al. [24], Staal et al. [23] and Soares et al. [28]. Table I summarizes the results of vessel segmentation for above mentioned methods. It shows the results in term of average accuracy and their standard deviation. Fig. 6 illustrates the blood vessel segmentation results for proposed method.

IV. CONCLUSION

The problem with retinal images is that the visibility of vascular pattern is usually not good. So, it is necessary to enhance the vascular pattern. In this paper, a wavelet based colored retinal image blood vessel segmentation technique is proposed. Vascular pattern is enhanced and sharpened prior to their detection. Vessel and located and segmentation mask is created using edge detection algorithm and morphological dilation operator respectively. We have tested our technique on publicly available STARE database of manually labeled images. Experimental results show that our method performs well in enhancing and segmenting the vascular pattern.

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TABLE I
VESSEL SEGMENTATION RESULTS (STARE IMAGES)

Segmentation Methods	Average Accuracy	Standard Deviation
2nd. Observer	0.9351	0.0171
Hoover et al.	0.9275	0.0247
Staal et al.	0.9516	0.0329
Soares et al.	0.9480	0.0298
Proposed Method	0.9439	0.0253

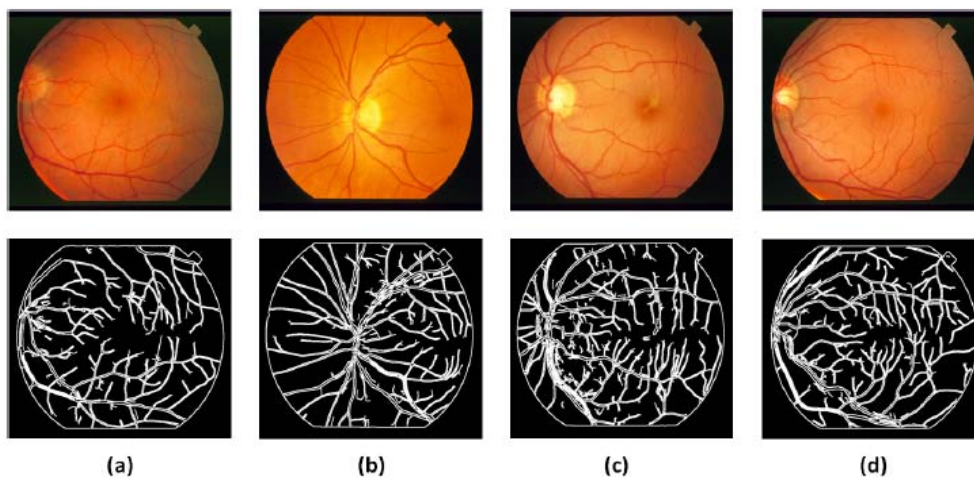


Fig. 6. Row 1: Original retinal images from STARE database, Row 2: Blood vessel segmentation mask for each retinal image